

# Harshavardhan Reddy Mulinti

## Machine Learning Engineer

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### PROFESSIONAL SUMMARY

Results-driven Machine Learning Engineer with 4+ years of experience designing and deploying production-ready ML systems across cloud and distributed environments. Deep expertise in deep learning, sequence modeling, reinforcement learning, NLP, and Generative AI (RAG, LLMs, Prompt Engineering). Proficient in Python, PyTorch, TensorFlow, and scikit-learn with extensive hands-on experience in AWS SageMaker, Azure ML, Kafka, Flink, and end-to-end MLOps (MLflow, Airflow, Docker, Kubernetes). Proven ability to build real-time inference pipelines, reduce latency, and architect scalable solutions that power recommendation engines, predictive modeling, and high-impact business applications.

### CORE ML COMPETENCIES

- Deep Learning: CNNs, RNNs, LSTMs, Transformers, BERT, Attention Mechanisms
- NLP & Generative AI: Hugging Face Transformers, LLMs, RAG Pipelines, Prompt Engineering, spaCy
- Reinforcement Learning: Reward modeling, policy optimization, LTV-driven RL frameworks
- Time Series & Forecasting: ARIMA, Prophet, LSTM-based temporal sequence modeling
- MLOps & Production ML: MLflow, Airflow, Kubeflow, CI/CD pipelines, model drift monitoring, A/B testing
- Vector Search & Embeddings: FAISS, Pinecone, embedding-driven candidate generation
- Cloud ML Platforms: AWS SageMaker, Azure ML, AWS EMR, Kubernetes (EKS/AKS)
- Feature Engineering: Real-time feature pipelines (Kafka, Flink), feature stores, EDA

### TECHNICAL SKILLS

**Programming & Scripting:** Python (NumPy, Pandas, scikit-learn, TensorFlow, PyTorch, Keras), R, Scala

**Machine Learning & AI:** Supervised & Unsupervised Learning, Deep Learning (CNNs, RNNs), NLP (BERT, spaCy, Transformers), Time Series Forecasting, Feature Engineering, Ensemble Methods (XGBoost, Random Forest, Gradient Boosting), Reinforcement Learning, Hyperparameter Optimization (Optuna, Ray Tune), Generative AI, Prompt Engineering, Retrieval-Augmented Generation (RAG)

**Cloud Platforms & Data Warehousing:** AWS (EC2, S3, Glue, Redshift, SageMaker, Lambda, Step Functions, EMR), Azure (Data Lake, Azure ML, AKS), Snowflake, GCP (BigQuery)

**MLOps & Deployment:** Docker, Kubernetes, Kubeflow, MLflow, Flask, FastAPI, Jenkins, Terraform, CI/CD, Model Monitoring, Model Versioning, TensorFlow Serving

**Data Engineering & Big Data:** Apache Spark, Hadoop, Kafka, Apache Flink, Airflow, Prefect, AWS EMR, Data Lakes, SQL (MySQL, PostgreSQL), NoSQL (MongoDB, DynamoDB)

**Data Analysis & Visualization:** EDA, A/B Testing, Statistical Analysis, Predictive Analytics, Data Wrangling, Tableau, Power BI, SSRS, Plotly, Matplotlib

**Tools & Environments:** Jupyter Notebook, VS Code, RStudio, Linux Shell, Git/GitHub, ELK Stack, RESTful Services, Postman, JIRA

### PROFESSIONAL EXPERIENCE

#### Netflix | Machine Learning Engineer | Los Angeles, CA

June 2024 – Present

- Designed and deployed a foundation-model-based sequence prediction system using Transformer architectures on AWS SageMaker and EC2 GPU clusters, improving 6-month member retention by 7% through accurate long-term viewing behavior modeling.
- Optimized real-time inference in PyTorch and TensorFlow Serving via key-value caching, model quantization, and distributed deployment on AWS EKS — reducing P99 latency by 30% and sustaining 2M+ requests/sec globally.
- Engineered a Reinforcement Learning reward optimization framework using AWS Lambda and Step Functions to balance short-term engagement with long-term subscriber LTV, delivering +5% gains in satisfaction and retention metrics.
- Built real-time feature pipelines with Apache Kafka and Apache Flink to process high-volume clickstream data, enabling sub-second model refreshes and improving real-time recommendation relevance by 12%.
- Architected and maintained end-to-end MLOps pipelines (Airflow, MLflow, Docker, Kubernetes) for model training, versioning, and production deployments — improving release reliability and deployment velocity.
- Collaborated with data and infrastructure teams to develop vector-based retrieval systems using FAISS and Pinecone, enhancing embedding-driven candidate generation and boosting cold-start recommendation accuracy by 10%.

- Engineered end-to-end ML solutions for finance and retail verticals, translating business requirements into scalable ML pipelines using Python, scikit-learn, and TensorFlow — covering data ingestion to production deployment.
- Built cloud-native ML infrastructure on Microsoft Azure (Azure Data Lake, Azure ML, AKS) to automate data ingestion, model training, and deployment for 20+ predictive models — reducing infrastructure costs by 12% and accelerating rollout by 20%.
- Developed RESTful APIs using Flask and FastAPI to operationalize ML models, reducing inference latency by 18% and seamlessly integrating with microservices and client-facing applications.
- Deployed optimized CNN and RNN models for computer vision and time-series forecasting — improving predictive accuracy by 15%, reducing false positives by 10%, and increasing inference throughput by 15%.
- Implemented robust MLOps and CI/CD pipelines using Azure DevOps, MLflow, and Git for continuous integration, automated retraining, reproducibility, and model drift monitoring across production environments.
- Created interactive Power BI dashboards to track ML model performance, business KPIs, and ROI metrics — enabling data-driven decision-making for senior stakeholders.

## **EDUCATION**

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**Master of Science in Data Science** | Rowan University, Glassboro, NJ, USA